

Hospital Cleaning Notebook

This notebook implements the requested cleaning pipeline, creates the plots in the same notebook, and saves `hospital_cleaned.csv`. Each code cell includes comments describing its purpose.

```
# Module 2: Data Cleaning & Transformation
## Hospital Operations & Patient Analytics Dashboard
```

```
import pandas as pd
import numpy as np
```

```
## Load Dataset
```

```
df = pd.read_csv("hospital_raw_data.csv")
```

```
print("Dataset Loaded Successfully")
```

Dataset Loaded Successfully

```
## Dataset Overview
```

```
print("Rows :", df.shape[0])
print("Columns :", df.shape[1])
print("Shape :", df.shape)
```

```
Rows : 10000
Columns : 61
Shape : (10000, 61)
```

```
df.head()
```

	Hospital ID	Hospital Name	Hospital Type	City
State \				
0	HOSP00016	C K Hospital	Private	Mumbai
Maharashtra				
1	HOSP00019	Cure Well Hospital	Private	Bengaluru
Karnataka				
2	HOSP00004	Yashoda Hospital	Private	Hyderabad
Telangana				
3	HOSP00017	Dharma Hospital	Private	Delhi
Delhi				
4	HOSP00017	Dharma Hospital	Private	Delhi
Delhi				

	Department	Department ID	Doctor	Nurses	Staff
Count ... \					
0	General Medicine	DEPT004	Dr. Arjun Sharma	12	
257 ...					

1	General Surgery	DEPT010	Dr. Sneha Rao	15
288	...			
2	Pulmonology	DEPT005	Dr. Priya Reddy	20
181	...			
3	Pulmonology	DEPT005	Dr. Rahul Verma	23
170	...			
4	Icu	DEPT009	Dr. Priya Reddy	34
116	...			

	Staff_Utilization_Calc	Readmission_Flag	Transferred_Flag	\
0	43.0	0	0	
1	49.4	0	0	
2	30.7	0	0	
3	28.5	0	1	
4	20.1	0	0	

	Equipment_InUse_Flag	Doctor Experience (Years)	Patient Satisfaction Score	\
0	0		3	
3.4				
1	0		24	
4.3				
2	1		20	
4.5				
3	0		14	
3.3				
4	1		13	
3.1				

	Ward Type	Follow-up Required	Payment Status	Discharge Status
0	General	No	Paid	Recovered
1	General	No	Paid	Recovered
2	General	No	Paid	Recovered
3	Private	No	Insurance	Recovered
4	Semi-Private	No	Paid	Recovered

[5 rows x 61 columns]

df.info()

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 10000 entries, 0 to 9999

Data columns (total 61 columns):

#	Column	Non-Null Count	Dtype
0	Hospital ID	10000 non-null	object
1	Hospital Name	10000 non-null	object
2	Hospital Type	10000 non-null	object
3	City	10000 non-null	object
4	State	10000 non-null	object

5	Department	10000	non-null	object
6	Department ID	10000	non-null	object
7	Doctor	10000	non-null	object
8	Nurses	10000	non-null	int64
9	Staff Count	10000	non-null	int64
10	Patient ID	10000	non-null	object
11	Patient Name	10000	non-null	object
12	Gender	10000	non-null	object
13	Age	10000	non-null	int64
14	Admission Date	10000	non-null	object
15	Discharge Date	10000	non-null	object
16	Diagnosis	10000	non-null	object
17	Treatment	10000	non-null	object
18	Medication	10000	non-null	object
19	Admission Type	10000	non-null	object
20	Test Result	10000	non-null	object
21	Blood Type	10000	non-null	object
22	Beds Available	10000	non-null	int64
23	ICU Beds	10000	non-null	int64
24	Bed Number	10000	non-null	int64
25	Bed Occupied	10000	non-null	object
26	Equipment	10000	non-null	object
27	Length of Stay	10000	non-null	int64
28	Room No	10000	non-null	int64
29	Billing Amount	10000	non-null	int64
30	Insurance Provider	10000	non-null	object
31	Readmission	10000	non-null	object
32	Dept_Bed_Capacity_Derived	10000	non-null	int64
33	Beds_Occupied_Count	10000	non-null	int64
34	Equipment ID	10000	non-null	object
35	Equipment Number	10000	non-null	object
36	Equipment Status	10000	non-null	object
37	Equipment_Total_Inventory	10000	non-null	int64
38	Equipment_Usage_Duration_Hours	10000	non-null	float64
39	Transferred	10000	non-null	object
40	Transfer_From_Department	10000	non-null	object
41	Transfer_To_Department	10000	non-null	object
42	Transfer_Date	1760	non-null	object
43	Number_of_Transfers	10000	non-null	int64
44	Dept_ICU_Bed_Capacity_Derived	10000	non-null	int64
45	Dept_Staff_Capacity_Derived	10000	non-null	int64
46	Admissions_Rate_%_Derived	10000	non-null	float64
47	Staff_Utilization_%_Derived	10000	non-null	float64
48	Bed_Occupancy_Rate_%	10000	non-null	float64
49	Bed_Occupancy_Rate_Calc	10000	non-null	float64
50	ICU_Occupancy_Rate_Calc	10000	non-null	float64
51	Staff_Utilization_Calc	10000	non-null	float64
52	Readmission_Flag	10000	non-null	int64
53	Transferred_Flag	10000	non-null	int64

```

54 Equipment_InUse_Flag      10000 non-null int64
55 Doctor Experience (Years)  10000 non-null int64
56 Patient Satisfaction Score 10000 non-null float64
57 Ward Type                  10000 non-null object
58 Follow-up Required         10000 non-null object
59 Payment Status             10000 non-null object
60 Discharge Status           10000 non-null object
dtypes: float64(8), int64(19), object(34)
memory usage: 4.7+ MB

```

Dataset Statistics

```
df.describe(include='all')
```

	Hospital ID	Hospital Name	Hospital Type	City
State \				
count	10000	10000	10000	10000
unique	20	20	2	6
top	HOSP00012	Medindia Hospital	Private	Chennai
Nadu				Tamil
freq	532	532	8496	1706
mean	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN
max	NaN	NaN	NaN	NaN

	Department	Department ID	Doctor	Nurses	Staff
Count \					
count	10000	10000	10000	10000.00000	
unique	10	10	4	NaN	
top	Neurology	DEPT002	Dr. Rahul Verma	NaN	
freq	1033	1033	2556	NaN	
mean	NaN	NaN	NaN	27.53610	

341.421000				
std	NaN	NaN	NaN	13.24947
149.201603				
min	NaN	NaN	NaN	5.000000
80.000000				
25%	NaN	NaN	NaN	16.000000
214.750000				
50%	NaN	NaN	NaN	28.000000
343.000000				
75%	NaN	NaN	NaN	39.000000
470.000000				
max	NaN	NaN	NaN	50.000000
600.000000				

... Staff_Utilization_Calc Readmission_Flag				
Transferred_Flag \				
count	...	10000.000000	10000.000000	10000.000000
unique	...	NaN	NaN	NaN
top	...	NaN	NaN	NaN
freq	...	NaN	NaN	NaN
mean	...	57.924860	0.498700	0.176000
std	...	25.296697	0.500023	0.380839
min	...	13.300000	0.000000	0.000000
25%	...	36.375000	0.000000	0.000000
50%	...	58.200000	0.000000	0.000000
75%	...	79.600000	1.000000	0.000000
max	...	100.000000	1.000000	1.000000

Equipment_InUse_Flag Doctor Experience (Years) \		
count	10000.000000	10000.000000
unique	NaN	NaN
top	NaN	NaN
freq	NaN	NaN
mean	0.335500	15.39290
std	0.472189	8.61157
min	0.000000	1.00000
25%	0.000000	8.00000
50%	0.000000	15.00000
75%	1.000000	23.00000

max	1.000000	30.00000
Patient Satisfaction Score Ward Type Follow-up Required Payment		
Status \		
count	10000.000000	10000
10000		
unique	NaN	4
3		
top	NaN	General
Paid		No
freq	NaN	3958
4435		
mean	4.000110	NaN
NaN		
std	0.575849	NaN
NaN		
min	3.000000	NaN
NaN		
25%	3.500000	NaN
NaN		
50%	4.000000	NaN
NaN		
75%	4.500000	NaN
NaN		
max	5.000000	NaN
NaN		

Discharge Status	
count	10000
unique	3
top	Recovered
freq	8269
mean	NaN
std	NaN
min	NaN
25%	NaN
50%	NaN
75%	NaN
max	NaN

[11 rows x 61 columns]

Check Missing Values

```
missing_values = df.isnull().sum()

print("Missing Values in Each Column")
print(missing_values)
```

```

Missing Values in Each Column
Hospital ID          0
Hospital Name        0
Hospital Type        0
City                 0
State                0
..
Patient Satisfaction Score  0
Ward Type                 0
Follow-up Required         0
Payment Status             0
Discharge Status           0
Length: 61, dtype: int64

df = df.fillna({
    "Age": df["Age"].median(),
    "Gender": df["Gender"].mode()[0],
    "Department": df["Department"].mode()[0]
})

print(df.isnull().sum())

Hospital ID          0
Hospital Name        0
Hospital Type        0
City                 0
State                0
..
Patient Satisfaction Score  0
Ward Type                 0
Follow-up Required         0
Payment Status             0
Discharge Status           0
Length: 61, dtype: int64

## Check Duplicate Records

duplicate_records = df.duplicated().sum()

print("Duplicate Records :", duplicate_records)

Duplicate Records : 0

## Remove Duplicate Records

df.drop_duplicates(inplace=True)

print("Duplicate records removed successfully.")

Duplicate records removed successfully.

## Standardize Department Names

```

```

df["Department"] = df["Department"].str.strip().str.title()
print(df["Department"].unique())

['General Medicine' 'General Surgery' 'Pulmonology' 'Icu' 'Oncology'
 'Neurology' 'Orthopedics' 'Cardiology' 'Nephrology' 'Psychiatry']

## Validate Cleaned Dataset

print("Rows :", df.shape[0])
print("Columns :", df.shape[1])
print("Missing Values")
print(df.isnull().sum())
print()
print("Duplicate Records :", df.duplicated().sum())

Rows : 10000
Columns : 61
Missing Values
Hospital ID          0
Hospital Name        0
Hospital Type        0
City                 0
State                0
..
Patient Satisfaction Score  0
Ward Type                   0
Follow-up Required          0
Payment Status              0
Discharge Status            0
Length: 61, dtype: int64

Duplicate Records : 0

## Export Cleaned Dataset

df.to_csv("hospital_cleaned.csv", index=False)
print("Cleaned dataset exported successfully.")
Cleaned dataset exported successfully.

df.head()

```

	Hospital ID	Hospital Name	Hospital Type	City
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0	0	3	
3.4			
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2	1	20	
4.5			
3	0	14	
3.3			
4	1	13	
3.1			

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[5 rows x 61 columns]